

AM234/BME234 Homework 1

Due: Wednesday April 22, 2026 at 11:59 pm

1. (30 pt) For each of the below questions, write your answer and provide a brief (2-3 sentence) explanation.

- (a) (5 pt) Consider an HMM defined over m states where each state emits one of k possible observations. How many parameters are needed to fully specify this model? (Assume that there are no begin or end states.)
- (b) (5 pt) True or False? Given a sequence $X = x_1, \dots, x_L$ and a state path $\pi = \pi_1, \dots, \pi_L$, we can compute the joint probability of a sequence and a state path by

$$P(X, \pi) = P(\pi_1)P(x_1)P(\pi_2 | \pi_1)P(x_2 | x_1) \dots P(\pi_L | \pi_{L-1})P(x_L | x_{L-1})$$

- (c) (5 pt) True or False? The following conditional independence statement holds in an HMM:

$$(x_n \perp\!\!\!\perp x_{n-10}) \mid \pi_{n-2}.$$

If the above notation is unclear, this statement reads that x_n and x_{n-10} are conditionally independent given π_{n-2} .

- (d) (5 pt) True or False? The following conditional independence statement holds in an HMM:

$$(\pi_n \perp\!\!\!\perp \pi_{n-10}) \mid x_{n-2}.$$

- (e) (10 pt) Let X be a sequence of length L where x_i denotes the state at position i . Assume that each x_i is drawn from the finite alphabet $\mathcal{X}$ and that none of these x_i corresponds to an end state. The probability of X under a Markov chain is given by the formula

$$P(X) = P(x_1, \dots, x_L) = P(x_1) \prod_{n=2}^L P(x_n | x_{n-1}).$$

Prove that the probability of observing all sequences of length L sums to 1. In other words,

$$\sum_{x_1} \sum_{x_2} \dots \sum_{x_L} P(x_1, x_2, \dots, x_L) = 1.$$

This result shows that a Markov chain defines a valid probability measure over all sequences of length L .

2. (30 pt) Suppose we are interested in discriminating promoter sequences from those that are not, which hereafter we refer to as *negative sequences*. To answer this question, we could learn two Markov chain models on promoter sequences and negative sequences. States in these Markov chain models are not individual nucleotides. Instead, they are overlapping k -mers (a k -mer is a substring of length k). For example, suppose we were given a sequence starting with “ACTGA...” and k equaled 3. Then, the first few states would be [ACT, CTG, TGA, ...]. Despite this increase in the dimensionality of each state, we can fit a Markov chain model the same way we did in class. Given these two Markov models, we could then classify a novel sequence X as being a promoter sequence if

$$\log \frac{P(X \mid \text{promoter sequence model})}{P(X \mid \text{negative sequence model})} > 0.$$

In this problem, we ask you to implement these Markov models and investigate the best choice of k .

- (a) (15 pt) Implement all functions in `problem_2.py`.
 - (b) (3 pt) Report the accuracy on the train set and the validation set when $k = 1$.
 - (c) (4 pt) Plot the validation accuracy for all values of k from 1 to 5, inclusive.
 - (d) (4 pt) What trends do you observe in your above plot? Why do you think those trends occur?
 - (e) (4 pt) List two problems of choosing a very large value of k (e.g. $k > 20$).
3. (20 pt) In this problem, you will explore the Variant Call Format (VCF)—a text file format used in computational biology for storing gene sequence variations. The VCF specification can be found at <https://samtools.github.io/hts-specs/VCFv4.1.pdf>.

Answer the following questions about the `chr21_filtered.vcf.gz` file. Please submit code for all subparts in a file named `problem_3.py`.

- (a) (4 pt) How many single-nucleotide polymorphisms (SNPs) are listed in this file? (Note that some VCF rows contain multiple alt alleles. Therefore, because a variant is a 4-tuple of (chrom, pos, ref, alt), one VCF row may contain multiple variants.)
- (b) (4 pt) Plot a histogram of variant allele frequencies. How many variants have an allele frequency less than 1%?
- (c) (4 pt) How many samples does this VCF file contain genotype data for? How many distinct variants does the average individual contain?
- (d) (4 pt) What is dbSNP? How many variants in this VCF file have a dbSNP id?
- (e) (4 pt) What is a Phred-quality score? Look this up. How accurate is a variant call if it has a Phred score of 45?